

Accurate FP64 Seismic Simulation on Modern NVIDIA Architectures: Co-designing Mixed Precision, Compressed Checkpointing, and Prefetch for PDE-Based Reverse Time Migration

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Abstract

Scientific simulations such as Reverse Time Migration (RTM) solve partial differential equations (PDEs) demanding FP64/FP32 accuracy, yet modern NVIDIA GPUs increasingly optimize for low-precision AI workloads (FP8/BF16) and expose new memory hierarchies, tensor cores, and dense linear-algebra libraries with mixed-precision extensions. Our prior work, **GPUZIP**, accelerated RTM on V100 systems by combining GPU-side lossy compression (NVIDIA Bitcomp via nvCOMP) with checkpoint prefetching, cutting host–GPU transfer volume by up to $80\times$ and delivering up to $5.1\times$ end-to-end speedup. This project extends GPUZIP to modern NVIDIA hardware over a 12-month programme in two stages. In the first six months, using H100 80 GB cloud GPUs, we will: (1) port Awave-3D and GPUZIP to the NVIDIA HPC SDK; and (2) re-evaluate compressed checkpointing and prefetch on Hopper memory, introducing **differential checkpointing**—a hybrid scheme that intermixes full wavefield snapshots with compressed inter-timestep deltas to reduce both transfer volume and recomputation cost. In the second six months, using an RTX PRO 6000 Blackwell GPU, we will: (3) introduce mixed-precision wavefield updates via cuBLAS mixed-precision GEMM and cuSOLVER iterative refinement; and (4) map where Ozaki-scheme FP64 emulation via low-precision tensor cores, cuFFT-accelerated spectral operators, and the precision $\times$ compression interaction improve a memory-bound stencil solver on AI-first hardware. Outcomes include an open-source GPUZIP v3.0 release and a peer-reviewed publication.

Keywords: high-performance computing; reverse time migration; mixed precision; lossy compression; checkpointing

NVIDIA Platforms

HPC SDK (nvc++/nvfortran, CUDA-X math libraries, Nsight Systems/Compute). *Experience:* team uses CUDA and Nsight Systems routinely. *Adoption:* migrate Awave-3D build from clang to nvc++; validate numerical parity against existing baselines. Nsight Compute provides kernel-level roofline analysis to identify memory-bandwidth-bound vs. compute-bound kernels and objectively scope tensor-core opportunities.

nvCOMP / Bitcomp (GPU-side lossy compressor). *Experience:* integrated and validated in GPUZIP v2.0; Bitcomp delivered the best accuracy-compression balance on V100. *Adoption:* re-tune error bounds for full-checkpoint and delta-checkpoint paths; re-benchmark throughput against HBM3 bandwidth on H100.

cuBLAS (mixed-precision extensions) and cuSOLVER (iterative-refinement solver). Both ship inside the HPC SDK. *Experience:* team has used BLAS and LAPACK routines in prior HPC

work. *Adoption*: apply cuBLAS mixed-precision GEMM (FP32 compute with FP64 accumulation) and cuSOLVER iterative refinement to dense sub-kernels in the RTM pipeline. Evaluate Ozaki-scheme FP64 emulation via low-precision tensor-core units on the RTX PRO 6000 Blackwell — a GPU whose negligible native FP64 makes it the ideal platform to characterise whether tensor-core emulation can substitute for hardware FP64 in scientific workloads.

cuFFT. *Experience*: used in prior spectral operator work. *Adoption*: benchmark cuFFT-based pseudo-spectral derivative operators as an alternative to FDM stencils for selected wavefield updates; characterise the memory-bandwidth vs. compute trade-off relative to the standard finite-difference path on H100 HBM3.

H100 80 GB SXM5 (cloud) and RTX PRO 6000 Blackwell (hardware). H100 is used for Phases 1–2 (FP64-heavy, multi-GPU HPC requiring HBM3 and NVLink). RTX PRO 6000 is used for Phases 3–4 (mixed-precision and Ozaki-scheme study, local development, and continuation beyond the cloud expiry window). The RTX PRO 6000 Blackwell’s 5th-generation tensor cores with FP8/FP4 support and 96 GB GDDR7 ECC memory make it a scientifically distinct platform for Ozaki-scheme emulation. Recent work has demonstrated up to $2.3\times$ speedup over native FP64 DGEMM on this specific GPU.

Dataset and Model

- **Size:** $\approx XTB$ total (confirm from REDU). Checkpoint pools reached ≈ 330 GB host RAM on V100; single Marmousi3D checkpoint ≈ 500 MB.
- **Data type:** seismic velocity models and traces (numerical grids).
- **Source:** Salt (SEG/EAGE) and Marmousi3D are open; Large dataset is in-house.
- **Confidentiality:** non-confidential, no personal data, no NDA.
- **AI model parameters:** not applicable (numerical PDE solver).
- **Storage location:** Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil; staged for upload to cloud.

Introduction

Hardware is trending toward low precision for AI, but PDE-based scientific simulation requires high precision to remain trustworthy. RTM, a seismic imaging method that solves the wave equation by finite differences over thousands of time steps, is a canonical example: geophysicists use the migrated image to guide expensive drilling decisions, so numerical fidelity is non-negotiable. RTM is also memory-intensive, requiring hundreds of gigabytes of wavefield state and relying on checkpointing to trade memory against recomputation in its forward/backward adjoint structure.

Our prior work addressed the dominant bottleneck (host–GPU checkpoint transfer over PCIe) with **GPUZIP** [1][2], which compresses checkpoints on the GPU (Bitcomp) and proactively prefetches them, reaching up to $5.1\times$ speedup on V100 while preserving image quality (SSIM ≈ 1 , near-zero relative error). Three shifts now motivate this project. First, **the memory bottleneck is changing**: Hopper-class coherent interconnects (HBM3, NVLink-C2C) shrink the PCIe wall, making it an open empirical question whether compressed checkpointing still pays and how the optimal cache size and prefetch schedule shift. Second, **checkpoint strategies can be extended**: rather than storing only full wavefield snapshots, a hybrid scheme can intermix full checkpoints with compressed inter-timestep deltas (small difference fields) that allow partial forward recomputation from the nearest full checkpoint without restarting from scratch,

reducing both storage volume and recomputation cost. Third, **modern math libraries reshape the precision/performance trade space**: cuBLAS exposes mixed-precision GEMM with FP64-accumulation paths, cuSOLVER offers iterative-refinement solvers, and the Ozaki scheme enables FP64-accurate DGEMM via low-precision tensor cores, an approach recently shown to achieve up to $2.3\times$ speedup over native FP64 DGEMM on the RTX PRO 6000 Blackwell [3]. To our knowledge, this is the first systematic co-design of differential checkpointing, lossy compression, mixed-precision linear algebra, and prefetch for an adjoint PDE solver on modern NVIDIA hardware.

Methods

Phase 1 — HPC SDK port and validation (M1–M2, H100). Build Awave-3D and GPUZIP with nvc++, re-integrate nvCOMP/Bitcomp, and reproduce V100 baselines on H100. Profile with Nsight Systems and Nsight Compute to establish roofline positions for all kernels, confirming that FDM stencils are memory-bandwidth-bound before any changes are attempted.

Phase 2 — Differential checkpointing and compression on Hopper (M2–M5, H100). Re-characterise the GPUZIP checkpoint pipeline on Hopper (HBM3 bandwidth, larger L2, NVLink) and introduce *differential checkpointing*: a hybrid strategy that stores full compressed snapshots at coarse intervals and compressed inter-timestep delta fields at finer granularity. During the backward pass, deltas are accumulated from the nearest full checkpoint, avoiding full forward recomputation for intermediate timesteps. We characterise the accuracy impact of applying lossy compression independently to both full checkpoints and deltas (PSNR, SSIM, mean relative error), sweep GPU Checkpoint Cache size and prefetch schedule, and determine the optimal full-to-delta checkpoint ratio across all three datasets and checkpointing algorithms (Revolve, zCut, Uniform).

Phase 3 — Mixed-precision wavefield updates (M7–M10, RTX PRO 6000). Introduce reduced-precision compute paths in the finite-difference stencil (FP32 update with FP64 accumulation). For dense sub-kernels identified in Phase 1, apply cuBLAS mixed-precision GEMM and cuSOLVER iterative refinement; evaluate Ozaki-scheme FP64 emulation via low-precision tensor-core units as an alternative. The RTX PRO 6000 Blackwell — with negligible native FP64 but abundant FP8/FP4 tensor-core throughput — is the ideal platform for this study: it forces the question of whether Ozaki-scheme emulation can fully substitute for hardware FP64. Bound all errors jointly (compression $\times$ precision interaction) against the FP64 baseline using PSNR, SSIM, and mean relative error.

Phase 4 — Spectral operators and final evaluation (M10–M12, RTX PRO 6000). Benchmark cuFFT-based pseudo-spectral derivative operators as an alternative to FDM stencils for selected wavefield updates. Integrate findings from Phases 2–3 into a unified GPUZIP v3.0 release, covering differential checkpointing, mixed-precision paths, and Ozaki-scheme emulation. Multi-node scaling evaluation (≤ 8 GPUs). The deliverable is a precise map of which NVIDIA library and algorithm paths improve a memory-bound adjoint PDE solver and which do not.

Evaluation. All experiments span three datasets (Large, Marmousi3D, Salt) and three checkpointing algorithms (Revolve, zCut, Uniform). Kernel-level profiling: Nsight Compute (roofline, memory-traffic breakdowns); timeline profiling: Nsight Systems. Implementation: C/C++ with CUDA under the HPC SDK; OpenMP Cluster for multi-node scaling.

Expected Results

Timeline (12 months: M1–M6 on H100 cloud, M7–M12 on RTX PRO 6000):

Month	Activity	Hardware	Phase
M1–M2	Port Awave-3D + GPUZIP to HPC SDK (nvc++); reproduce V100 baselines; Nsight Compute roofline profiling.	H100 (ramp-up)	1
M2–M5	Differential checkpointing + lossy compression on Hopper; prefetch schedule optimisation; accuracy study (PSNR/SSIM/MRE).	H100 (peak)	2
M5–M6	Multi-node scaling (≤ 8 GPUs); open-source GPUZIP v3.0-alpha release; reproducibility package (REDU).	H100 (8 GPUs)	2
M7–M10	Mixed-precision stencil updates (cuBLAS/cuSOLVER/Ozaki); FP64 emulation on RTX PRO 6000; precision $\times$ compression interaction.	RTX PRO 6000	3
M10–M12	cuFFT spectral-operator benchmark; unified GPUZIP v3.0 release; analysis, write-up, and paper submission.	RTX PRO 6000	4

Projected outcomes: open-source GPUZIP v3.0 targeting Hopper and Blackwell NVIDIA architectures (MIT license, GitHub); peer-reviewed publication (target: SBAC-PAD / Euro-Par / IJH-PCA); public reproducibility package (datasets, Nsight profiles, quality metrics) via UNICAMP REDU.

Dependencies: the project is not conditional on this award being combined with another grant. Existing institutional support funds personnel and local infrastructure; none duplicates the requested NVIDIA cloud compute or hardware.

Project Support Details

We request both cloud H100 80 GB hours (Phases 1–2) and one RTX PRO 6000 Blackwell GPU (Phases 3–4). The H100 is required for FP64-heavy multi-GPU experiments demanding Hopper’s HBM3 bandwidth, FP64 Tensor Core throughput, and NVLink fabric. The RTX PRO 6000 is requested for the mixed-precision and Ozaki-scheme study in the second six months: its abundant FP8/FP4 tensor-core throughput and 96 GB GDDR7 ECC memory make it a scientifically distinct target for FP64 emulation research, and it provides local compute beyond the cloud expiry window.

Resource	Request	Cap
Cloud GPU hours (H100 80 GB)	~17,000 (Phases 1–2)	30,000
Concurrent GPUs	up to 8	8
Cloud storage	~4 TB	32 TB
Physical hardware	1$\times$ RTX PRO 6000 Blackwell (Phases 3–4)	≤ 8 GPUs

GPU-hour breakdown (placeholders — replace with measured pilot timings before submission):

- **Phase 1 port + roofline profiling:** ~2,000 GPU-hrs (H100)
- **Phase 2 differential checkpointing + compression study:** ~13,000 GPU-hrs (H100, bulk)
- **Profiling overhead (Nsight Compute sweeps):** ~2,000 GPU-hrs (H100)

Total $\approx$ 17,000 H100 hours, well under the 30,000-hour cap. Phases 3–4 run on the RTX PRO 6000 (local).

Cloud Readiness

Development environment is Linux (Ubuntu); Awave-3D and GPUZIP already build and run on Ubuntu, matching the cloud servers.

- **Importing data:** pull datasets from UNICAMP REDU on day one. (Experienced.)
- **Installing dependencies:** HPC SDK (nvc++), nvCOMP, cuBLAS, cuSOLVER, cuFFT — all via NVIDIA NGC containers. (Experienced.)
- **Building/running the codebase:** existing CMake build scripts and CLI workflow. (Experienced.)
- **Backing up results:** Nsight profiles and checkpoint outputs synced off ephemeral instances regularly. (Experienced.)
- **CUDA and NVIDIA SDKs:** HPC SDK + CUDA-X + nvCOMP integration is core to the project. (Experienced.)
- **Containers:** containerised builds from NVIDIA NGC base images for full reproducibility. (Experienced.)

Appendix B — CV(s) (required, 1 page per collaborator)

Appendix C — Citations